RP: Auditing Process Knowledge for Parsimonious and Explainable Industrial Cyberattack Detection

Anonymous submission

Abstract

Industrial Control Systems (ICSs) require attack detection that captures temporal behavior while remaining interpretable to operators. Engineering-informed detectors often incorporate physically plausible process rules, but such rules may be unreliable, redundant, or unhelpful for detection. We therefore study a different question: which engineering relationships should actually be admitted into a detector? Using the Secure Water Treatment (SWaT) dataset, we audit 27 candidate process relationships against training and validation data. Six relationships that pass all audit checks form a conservative CORE representation, while 16 are retained overall. We combine CORE with a 25-variable LSTM and evaluate it on a fixed chronological test partition across five random seeds. CORE achieves the highest mean test MCC (0.247) and F1 (0.161), compared with 0.183 and 0.137 for the LSTM alone. Using all 16 retained relationships performs worse than CORE, suggesting that more process knowledge does not necessarily improve detection. However, all models perform substantially worse on the chronological test data than on validation data, revealing a major limitation in generalization over time. Our contribution is a systematic way to audit engineering knowledge before incorporating it into a hybrid detector, favoring a smaller and more interpretable representation.

Code and reproducibility materials are available at: https://anonymous.4open.science/r/swat-symbolic-fusion-E31E/

Introduction

Industrial Control Systems (ICSs) integrate sensors, actuators, programmable logic controllers (PLCs), and supervisory computers through networked communication. As these mission-critical systems become increasingly connected, their attack surface expands, while successful cyberattacks can have direct physical consequences (Cárdenas, Amin, and Sastry 2008). This growing security concern has motivated the development of realistic testbeds for studying cyberattacks and evaluating detection methods. Among them, the Secure Water Treatment (SWaT) testbed models a realistic six-stage water-treatment process and has become widely used for evaluating industrial cyberattack detection methods (Mathur and Tippenhauer 2016; Goh et al. 2017a,b; Adepu and Mathur 2018). Detection in such cyber-physical environments presents a dual challenge: models must capture multivariate temporal behavior while providing evidence that operators can relate to the underlying physical process.

Existing ICS attack-detection methods range from data-driven temporal machine learning models to approaches that incorporate explicit process rules. While the latter can provide interpretable physical evidence, a key question remains: which engineering relationships are sufficiently reliable and informative to be incorporated into a detector? We address this question by combining a Long Short-Term Memory (LSTM) network that learns temporal process patterns with an audited, parsimonious representation of engineering knowledge, allowing cyberattacks to be detected using both learned temporal behavior and explicit process inconsistencies. To construct this representation, we draw on a broad set of engineering relationships identified with input from a domain expert and systematically audit them to determine which are sufficiently reliable to retain.

Our main contributions are:

- **Audit-driven hybrid detection:** We introduce a hybrid framework that combines temporal learning with a parsimonious, empirically audited set of engineering relationships, rather than assuming that more process knowledge is necessarily better.
- **Systematic knowledge audit:** We develop an empirical procedure for determining which candidate engineering relationships are sufficiently supported, reliable, and non-redundant for inclusion in the detector.

Related Work

One major line of ICS attack-detection research uses explicit knowledge of the physical process. Rule-based approaches encode expected relationships among sensors, actuators, and process states, allowing violations to provide interpretable evidence of abnormal behavior (Kang et al. 2016; Lin et al. 2018; Umer et al. 2020; Das, Adepu, and Zhou 2020; Al-Dhaheri, Zhang, and Mikhaylenko 2022; Cho 2026). These approaches show the value of engineering knowledge, but they require decisions about which process relationships, models, and thresholds should be used. Such decisions can require substantial manual effort, and physically plausible re-

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lationships may not remain reliable across different operating conditions.

Data-driven approaches instead learn patterns directly from multivariate ICS observations. Recurrent, convolutional, autoencoder, sequence-learning, and graph-based models have been used to capture temporal and cross-variable dependencies (Goh et al. 2017b; Inoue et al. 2017; Kravchik and Shabtai 2018; Kim, Yun, and Kim 2020; Audibert et al. 2020; Deng and Hooi 2021; Li et al. 2021; Tang et al. 2023). Feature-selection and dimensionality-reduction methods can further reduce the complexity of raw process observations (Bozdal, Ileri, and Ozkahraman 2024). These methods reduce dependence on manually specified process models, but their predictions do not necessarily reveal which recognizable physical relationship has been violated.

Recent hybrid approaches combine engineering knowledge with learned models to benefit from both physical interpretability and data-driven learning (Raman and Mathur 2022b; Sung, Raman, and Mathur 2022; Raman and Mathur 2022a, 2024; Rajalakshmi and Velmurugan 2026). Post-hoc attribution provides another way to interpret learned detectors (Fung, Zeng, and Bauer 2024). However, an important question remains: *which engineering relationships should be included in a hybrid detector in the first place?* Prior work has not generally treated this selection as a systematic empirical audit problem. A physically plausible relationship may still be rarely observed, unstable over time, or redundant with other evidence. Our work addresses this gap by auditing candidate engineering relationships against observed process behavior before incorporating them into the detector, producing a compact CORE representation that is both physically interpretable and empirically supported.

Problem Formulation and Method Overview

Problem Formulation

At each time step t, the system records a vector $\mathbf{x}_t \in R^d$ containing d sensor and actuator measurements, together with a binary label $y_t \in \{0, 1\}$, where 1 denotes attack and 0 normal operation. Rather than classifying each observation independently, the detector uses the most recent L observations:

$$\mathbf{X}_t = [\mathbf{x}_{t-L+1}, \ldots, \mathbf{x}_t] \in R^{L \times d}, \tag{1}$$

where L is the sequence length and d is the number of process variables. In our experiments, $L = 120$, corresponding to approximately 120 seconds of SWaT data. The goal is to estimate

$$P(y_t = 1 \mid \mathbf{X}_t), \tag{2}$$

the probability of an attack at time t.

Audited Process Knowledge and Hybrid Detection

The framework has two main branches: an LSTM branch that learns temporal behavior from selected process variables, and a symbolic branch that audits and scores engineering relationships. The two representations are then combined for attack classification, as illustrated in Figure 1.

First, we rank the 51 physical variables in the SWaT dataset using Random Forest importance computed from the TRAIN partition. Let $\mathcal{I} \subseteq \{1, \ldots, d\}$ denote the indices of the selected K variables. The reduced temporal input is

$$\widetilde{\mathbf{X}}_t = \mathbf{X}_t[:, \mathcal{I}] \in R^{L \times K}. \tag{3}$$

Thus, $\widetilde{\mathbf{X}}_t$ contains the same L time steps as $\mathbf{X}_t$, but only the K selected variables.

We use a RandomForestClassifier with 500 trees, class-balanced weights, and random seed 42, with all other hyperparameters set to the scikit-learn defaults. Variables are ranked by mean decrease in impurity (MDI) using the unscaled TRAIN data. We evaluate $K \in \{5, 10, 15, 20, 25, 30\}$ using a compact LSTM proxy on VALIDATION and select $K = 25$ based on MCC, with F1 and smaller K used as tie-breakers. The TEST partition is not used for ranking or subset selection.

Second, we define candidate engineering relationships based on the known behavior of the water-treatment process. Examples include valve–flow consistency, pump–level dynamics, coupled actuator states, and agreement between related sensors. Let r_j denote the jth candidate relationship, where $j = 1, \ldots, J$ and J is the total number of candidates.

For each relationship r_j, we compute a symbolic score from the process observations:

$$s_j(\mathbf{X}_t) \in R. \tag{4}$$

Here, $s_j(\mathbf{X}_t)$ measures how much the observed behavior deviates from the expected relationship r_j; a larger score indicates a greater deviation.

We then audit each candidate relationship using TRAIN and VALIDATION data for empirical support, temporal coverage, robustness, and redundancy. The audit retains 16 of the 27 candidate relationships. Among them, six relationships that pass all audit checks without identified caveats form the conservative CORE set $\mathcal{C}$. Their symbolic scores form the vector

$$\mathbf{s}_t = [s_j(\mathbf{X}_t)]_{j \in \mathcal{C}}. \tag{5}$$

Third, the LSTM processes the reduced observation window $\widetilde{\mathbf{X}}_t$ and produces a temporal representation:

$$\mathbf{h}_t = \mathrm{LSTM}(\widetilde{\mathbf{X}}_t), \tag{6}$$

where $\mathbf{h}_t \in R^{64}$ is the 64-dimensional hidden representation summarizing the recent temporal behavior of the selected process variables.

We then combine this learned representation with the audited symbolic scores:

$$\mathbf{z}_t = [\mathbf{h}_t \| \mathbf{s}_t], \tag{7}$$

where $\|$ denotes vector concatenation. Thus, $\mathbf{z}_t$ contains both the LSTM representation and the symbolic deviation scores.

Finally, a linear classification layer followed by a sigmoid function produces the attack probability:

$$\hat{p}_t = \sigma(\mathbf{w}^\top \mathbf{z}_t + b), \tag{8}$$

where $\mathbf{w}$ and b are learned parameters and $\sigma(\cdot)$ maps the output to a value between 0 and 1. The model is trained using binary cross-entropy on the TRAIN sequences.

In this framework, the LSTM represents temporal behavior learned from data, while the symbolic vector represents deviations from audited engineering relationships. The next section explains how these symbolic features are constructed and audited to form the CORE representation.

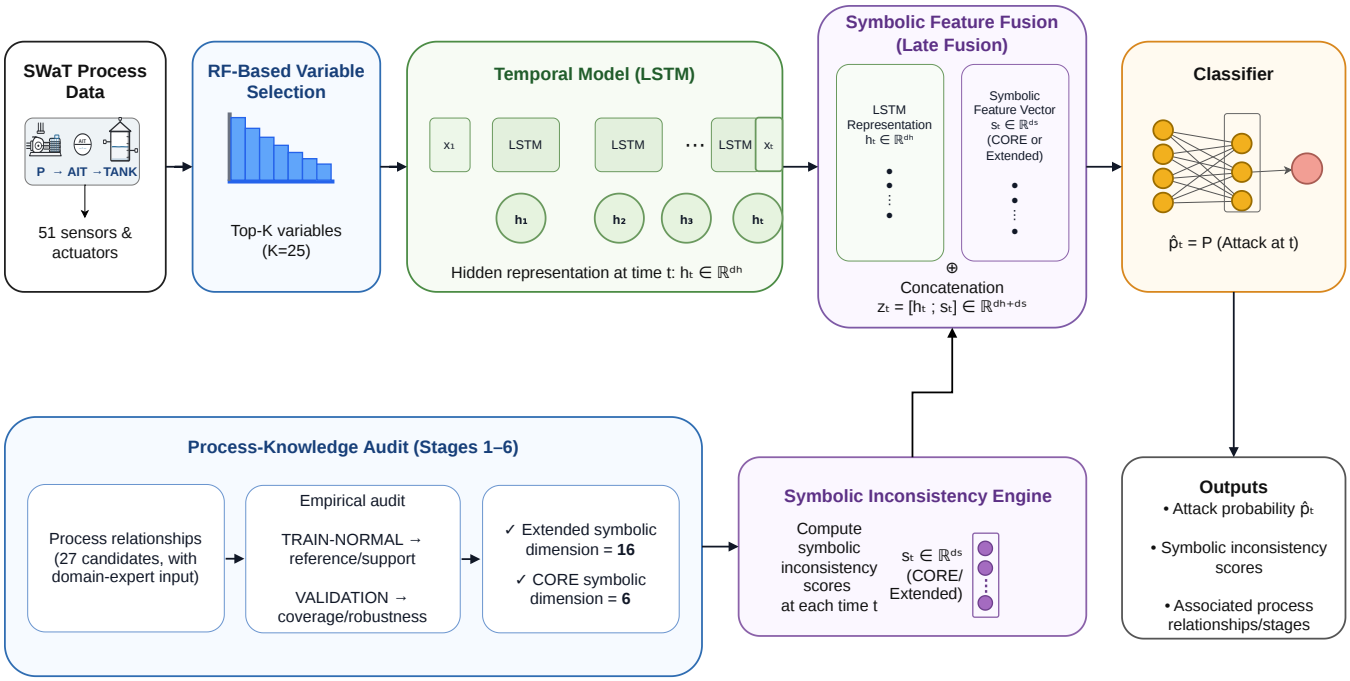


Figure 1: Overview of the proposed process-knowledge audit and hybrid detection framework. Audited symbolic features are combined with the LSTM temporal representation before attack classification.

Audited Symbolic Feature Construction

We begin with 27 candidate engineering relationships covering the six stages of SWaT, identified from the process structure with input from a domain expert. The candidates describe specific relationships among sensors and actuators, including valve–flow responses, pump–level dynamics, actuator coupling, and sensor–sensor consistency. These candidates do not all represent the same type of process knowledge: some reflect constraints enforced by control logic, others follow directly from physical process behavior, and others represent process responses whose timing or magnitude must be empirically calibrated from observed data. We convert each candidate relationship into a numerical symbolic score and then audit it using TRAIN and VALIDATION data to determine whether it should be retained.

For a candidate relationship r_j, the resulting score $s_j(\mathbf{X}_t)$ measures its deviation from normal process behavior, with larger values indicating greater deviation. Depending on the relationship, the score captures state mismatch, dynamic deviation, actuator coupling, or sensor–sensor residual behavior.

Symbolic Feature Audit

We audit each candidate relationship in three steps: empirical support, VALIDATION coverage, and robustness and redundancy checks. Only TRAIN and VALIDATION data are used; TEST is never used for symbolic construction or selection.

First, empirical support is evaluated using relation-specific effect-size, association, and minimum-sample criteria. Actuator–sensor state relations require a standardized mean difference (SMD) greater than 1.0 in magnitude and interquartile-range (IQR) overlap < 0.25; pump–slope relations require $|\mathrm{SMD}| > 0.5$ and IQR overlap < 0.5. Unconditional mass-balance/slope relations require an absolute correlation coefficient $|\rho| > 0.2$, $p < 0.01$, and explained variance > 0.04; sensor–sensor relations require a coefficient of variation (CV) < 0.5, $|\rho| > 0.3$, and $p < 0.01$. Actuator coupling requires a coupling rate ≥ 0.8, and Stage 2 event relations additionally require expected-direction consistency $\geq 95\%$.

Second, supported candidates must achieve VALIDATION coverage $\geq 90\%$ and a VAL-NORMAL mean score < 0.90; otherwise they are labeled LIMITED.

Third, we further check whether each retained relationship remains stable from TRAIN to VALIDATION, whether its normal reference data are sufficiently represented, whether it provides information distinct from other symbolic features, and whether its estimated time lag lies within the tested range rather than at its boundary.

Starting from 27 candidate relationships, the audit retains 16: six are selected as CORE and ten are retained with caveats. Two additional relationships pass the empirical-support criteria but fail the VALIDATION coverage requirement and are therefore labeled LIMITED, while nine fail the support criteria and are rejected. The resulting six CORE relationships are used as the symbolic representation in our primary hybrid detector. Detailed evidence for each candidate is provided in the supplementary material[1].

[1] https://anonymous.4open.science/r/swat-symbolic-fusion-E31E/Supplementary.pdf

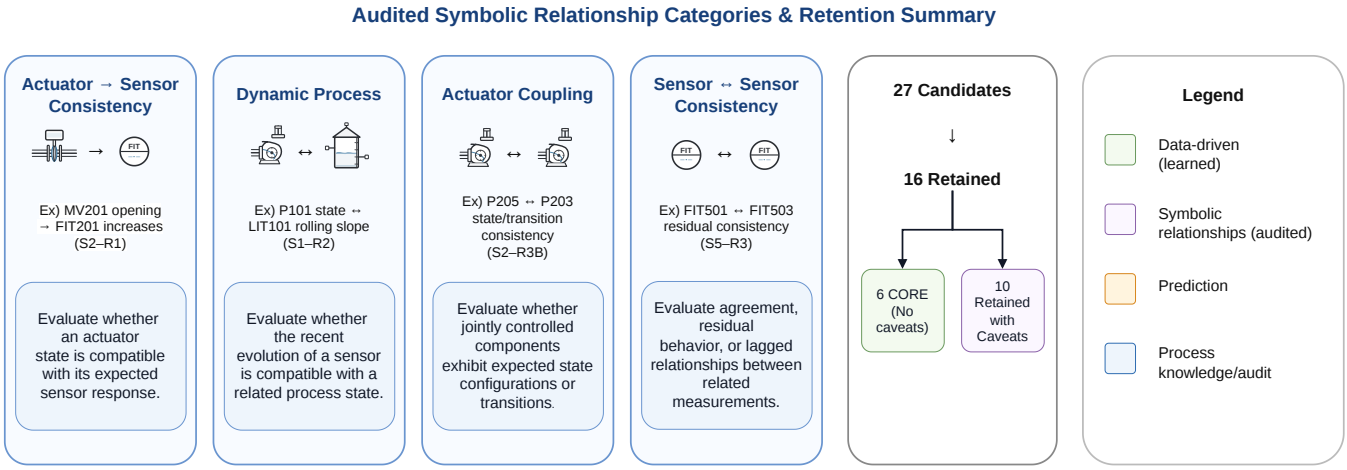


Figure 2: Audited symbolic relationship categories and retention summary. Representative process relationships include actuator–sensor consistency, dynamic process behavior, actuator coupling, and sensor–sensor consistency. The audit reduces 27 candidate relationships to 16 retained features, consisting of six CORE features and ten features retained with caveats.

CORE and Extended Symbolic Representations

Based on the audit, we construct two symbolic representations for model fusion. CORE contains the six retained relationships with no identified reliability concerns, whereas Extended contains all 16 retained relationships, including the ten with documented caveats. Thus, Extended contains CORE plus the ten additional caveated relationships.

The six CORE relationships are S1-R1 (MV101–FIT101 valve–flow consistency), S1-R2 (P101–LIT101 pump–level dynamics), S1-R4 (FIT101–LIT101 flow–level dynamics), S2-R1 (MV201–FIT201 valve–flow consistency), S2-R3B (P205–P203 actuator coupling), and S5-R3 (FIT501–FIT503 sensor–sensor residual consistency).

For model fusion, CORE provides a six-dimensional symbolic vector $\mathbf{s}_t$, while Extended provides a 16-dimensional vector. In both cases, the symbolic vector at the sequence's final time step is concatenated with the 64-dimensional LSTM representation $\mathbf{h}_t$ before classification. Comparing the two configurations allows us to test whether adding the ten caveated relationships improves attack detection over using only the six CORE relationships.

Experimental Setup

Dataset and Preprocessing

We evaluate the proposed framework on the SWaT dataset, a six-stage water-treatment testbed with 51 physical sensor and actuator variables (Goh et al. 2017a). The six stages correspond to raw-water intake and storage (S1), chemical dosing (S2), ultrafiltration (S3), dechlorination (S4), reverse osmosis (S5), and final product and backwash-related flows (S6). SWaT tag prefixes indicate the physical role of each variable: FIT denotes flow-indicator transmitters, LIT level-indicator transmitters, AIT analyzer-indicator transmitters, MV motorized valves, and P pumps.

SWaT provides a separate normal-operation collection (496,800 observations) and an attack collection (449,919 observations). The attack collection contains both normal and attack periods, with 54,621 attack-labeled observations grouped into 35 distinct attack events. Our detection experiments use this attack collection, which is divided chronologically into TRAIN, VALIDATION, and TEST partitions; the separate normal-only collection is not merged into these partitions.

Before modeling, we clean column names, timestamps, and binary labels. Timestamps and labels are then removed from the model inputs, leaving the 51 physical variables. Standardization parameters are estimated from TRAIN and applied unchanged to VALIDATION and TEST.

Chronological Evaluation Protocol

To prevent observations from the same attack event from appearing in different partitions, we do not randomly split individual observations. Instead, we divide the attack collection chronologically into TRAIN, VALIDATION, and TEST, with each split boundary placed during a normal operating period. This ensures that a single attack event is never divided across two partitions. TRAIN contains the earliest 270,000 observations and 22 complete attack events, followed by VALIDATION with 68,319 observations and 3 complete attack events, and TEST with the final 111,600 observations and 10 complete attack events.

The TEST partition is reserved exclusively for final evaluation. It is not used to estimate standardization parameters, rank variables with Random Forest, choose the number of selected variables, construct or audit symbolic features, tune model hyperparameters, or select the classification threshold. The separate normal-only SWaT collection is also excluded from these partitions because it was recorded during a different collection period; combining it with the attack collection could allow the model to learn differences between collection periods rather than differences between normal and attack behavior.

Comparison Models

We compare four primary configurations. **Full LSTM** uses all 51 physical variables, while **Selected LSTM** uses the 25 variables selected by Random Forest. **Neuro-Symbolic (NS) CORE** combines the Selected LSTM representation with the six CORE symbolic features, whereas **NS Extended** combines it with all 16 retained symbolic features. Comparing Selected LSTM with NS CORE evaluates whether adding the six CORE symbolic features improves attack detection, while comparing NS CORE with NS Extended tests whether adding the ten retained features with caveats provides additional benefit.

Model Training and Sequence Construction

LSTM sequences contain $L = 120$ consecutive observations, corresponding to approximately 120 seconds of SWaT data. We use stride 1, so the input window moves forward by one observation at a time, and the label of the final observation is used as the prediction target. Sequences are constructed separately within TRAIN, VALIDATION, and TEST, so no window crosses a partition boundary.

All LSTM configurations use one recurrent layer with 64 hidden units, followed by dropout 0.2 before the linear classification layer. We use Adam with learning rate 10^{-3}, batch size 256, and a maximum of 30 epochs. Because attack observations are less frequent than normal observations, training uses weighted binary cross-entropy with a positive-class weight calculated from TRAIN. Training stops if the VALIDATION loss does not improve for five consecutive epochs, and the best checkpoint is retained. To account for randomness in model training, each neural configuration is repeated using five fixed random seeds $\{7, 21, 42, 84, 123\}$.

After training, each model converts its predicted attack probability into a binary Normal/Attack decision using a threshold selected on VALIDATION. Among the unique predicted probabilities on VALIDATION, the threshold is chosen to maximize the Matthews Correlation Coefficient (MCC), which evaluates binary classification using both positive and negative predictions and is suitable for imbalanced data. Ties are resolved by higher F1, lower false-positive rate, proximity to 0.5, and then higher threshold. Once selected, the threshold is fixed and applied unchanged to TEST.

Evaluation Measures

We evaluate detection performance using MCC, F1, ROC-AUC, and inference latency. MCC measures overall binary classification quality and is particularly useful for imbalanced data. F1 balances precision and recall, while ROC-AUC measures how well the model separates attack from normal observations across different classification thresholds. Inference latency measures the time required to produce a prediction. All final performance results are reported on the fixed chronological TEST partition, which remains held out during model development.

Operator-Facing Explanation

Each retained symbolic feature corresponds to a named process relationship and SWaT stage. An alert can therefore report the predicted attack probability together with the highest-scoring symbolic relationships and their associated stages. This provides operators with interpretable physical evidence alongside the LSTM prediction. However, a high symbolic score only indicates that an audited process relationship is behaving abnormally; it does not determine whether the cause is a cyberattack, equipment fault, sensor failure, or another abnormal condition.

Table 1: Symbolic-feature screening across SWaT stages. Cand. = candidate relations; Ret. = retained relations; Lim. = limited relations; Rej. = rejected relations; Cav. = retained relations with documented caveats.

Stage	Cand.	Ret.	Lim.	Rej.	CORE	Cav.
S1	5	4	0	1	3	1
S2	4	2	2	0	2	0
S3	6	3	0	3	0	3
S4	6	1	0	5	0	1
S5	5	5	0	0	1	4
S6	1	1	0	0	0	1
Total	**27**	**16**	**2**	**9**	**6**	**10**

Results

Audited Symbolic Feature Screening

Of 27 candidate relationships, the audit retains 16, classifies two as LIMITED, and rejects nine. Six of the retained relationships have no identified audit caveats and form the conservative CORE set (Table 1).

The two LIMITED relationships, Stage 2 P205–AIT203 and P203–AIT202, show the expected response direction but insufficient temporal coverage. The six CORE relationships span actuator–sensor, dynamic-process, actuator-coupling, and sensor–sensor relationships (Figure 2).

Raw-Variable Selection

To select the number of physical variables, we evaluate RF-ranked subsets $K \in \{5, 10, 15, 20, 25, 30\}$ using a compact LSTM proxy on VALIDATION. The proxy uses 32 hidden units, at most 15 epochs, patience 3, and a fixed decision threshold of 0.5. We select the subset with the highest MCC, followed by higher F1 and then smaller K in case of ties.

This procedure selects $K = 25$, with VALIDATION MCC $= 0.064$ and F1 $= 0.065$. These 25 variables account for 94.59% of the cumulative RF importance. The proxy is used only to select the subset size and does not represent final detection performance.

Held-Out Detection Performance

Table 2 reports performance on the fixed chronological TEST partition. All model-development choices are fixed using TRAIN and VALIDATION before TEST evaluation. Results are mean±std across five fixed random seeds.

NS CORE achieves the highest mean TEST MCC (0.247) and F1 (0.161), while Full LSTM achieves the highest mean AUC (0.621). Relative to Selected LSTM, CORE improves

Table 2: Final held-out TEST performance (mean±std over five random seeds).

Model	Raw	Sym.	MCC	F1	AUC
Full LSTM	51	0	.135±.082	.066±.043	**.621±.084**
Selected LSTM	25	0	.183±.101	.137±.111	.520±.049
NS CORE	25	6	**.247±.063**	**.161±.063**	.600±.052
NS Extended	25	16	.155±.123	.084±.098	.577±.041

Table 3: Chronological generalization from VALIDATION to TEST. Values are means over five random seeds.

Configuration	VAL MCC	TEST MCC
Full LSTM	.821	.135
Selected LSTM	.820	.183
NS CORE	**.821**	**.247**
NS Extended	.819	.155

mean MCC, F1, and AUC, although none of the paired differences is statistically significant across five random seeds ($p \geq 0.0625$).

NS Extended performs worse than CORE in mean MCC and F1, indicating that adding all ten caveated relationships does not improve held-out detection. A size-matched six-feature control also achieves lower mean MCC (0.166), F1 (0.127), and AUC (0.529) than CORE, suggesting that CORE's result is not explained solely by using fewer symbolic features; however, these differences are also not statistically significant.

Robustness and Symbolic-Alignment Ablation

Performance drops substantially from VALIDATION to the later chronological TEST period, with mean MCC decreasing from approximately 0.82 to 0.135–0.247. This degradation is driven primarily by missed attacks rather than false alarms. In VALIDATION, one attack interval accounts for 70.6% of attack observations, and all four models detect approximately 96% of it. In TEST, six of ten attack intervals, comprising 78.4% of attack observations, are missed by all four models. Accordingly, attack recall falls from approximately 68% to 4–9%, while false-positive rates remain below 0.4%.

The shuffled-symbolic control shows little effect on VALIDATION: CORE differs from Shuffled NS CORE by only $+0.0006$ MCC, $+0.0005$ F1, and -0.0248 AUC. Thus, we find no consistent evidence that symbolic temporal alignment improves performance under the present late-fusion architecture.

Parsimony and Explainability

CORE provides a more parsimonious symbolic representation than Extended, using six rather than 16 relationships while achieving higher mean TEST MCC and F1. Its main benefit is not computational speed, but reducing the number of engineering relationships used by the detector without sacrificing detection performance. Consistent with this interpretation, mean TEST inference latency is similar across the neural models (0.064–0.079 ms/sequence).

The six CORE features also provide explicit process-level evidence. Each corresponds to a named physical relationship and SWaT stage, such as MV201–FIT201 valve–flow consistency or P205–P203 actuator coupling. When a CORE score is elevated, an operator can identify which relationship is behaving abnormally. This provides interpretable physical evidence alongside the LSTM prediction.

Discussion and Limitations

The results support the value of auditing engineering knowledge before incorporating it into a hybrid detector. Of 27 candidate relationships, 16 are retained and only six form the conservative CORE set, showing that physically plausible relationships are not necessarily reliable or useful for detection. CORE achieves the strongest mean held-out MCC and F1, with higher mean values than both the 16-feature Extended representation and a size-matched six-feature control. This suggests that simply adding more process knowledge does not improve detection and that the specific relationships selected through the audit may matter. However, these differences are not statistically significant. With five paired seeds, the exact two-sided Wilcoxon test has a minimum attainable p-value of 0.0625, limiting statistical power.

The CORE representation also improves parsimony while preserving process-level interpretability through six named engineering relationships. However, symbolic deviations should be interpreted as evidence of abnormal process behavior rather than as root-cause diagnoses. Symbolic normal references reflect training-observed operating modes rather than universal normal behavior; less frequent legitimate modes may therefore require mode-specific references.

The main limitation is chronological generalization. Mean MCC falls from approximately 0.82 on VALIDATION to 0.135–0.247 on TEST, primarily because later attack intervals are missed. This finding is consistent with recent work emphasizing that event-level detection and missed-incident rates can reveal operational weaknesses not captured by aggregate detection metrics (Heydari and Nyarko 2026). Although CORE achieves the highest mean TEST MCC of 0.247, its absolute performance remains weak and should not be interpreted as deployment-ready. Moreover, the experiments are limited to SWaT, so the audit and CORE representation should be evaluated on additional industrial processes before broader generalization can be established. Future evaluation should account for attack severity and measure detection delay relative to each event's physical timescale.

Conclusion

We presented a systematic audit for selecting engineering relationships for hybrid industrial cyberattack detection. On SWaT, the six-feature CORE achieves the strongest mean held-out MCC and F1 among the primary configurations, while adding more retained relationships provides no improvement. These results support careful selection of process knowledge, while the substantial chronological performance drop highlights generalization as an open challenge.

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